MODERNIZING DATA-DELIVERY TECHNOLOGIES
Combining Data-User Feedback with User-Support Expertise for Successful Data Tools

And Pivoting when Technology Changes!

Digital Data Products: Two Key Questions

1. What problem(s) need to be solved?
2. How do **users** want to solve those problems?

Unlocking the potential of data tools requires: understanding user needs (talking directly with data users); leveraging existing user support expertise; the pivoting unapologetically if user demands and tech developments shift.

Why This Matters to You

Elevate your potential to gather effective user feedback when developing digital data products and data-related services:

- When resources or time are in short supply
- That goes beyond "textbook" or "Google-search" research approaches used in ideal conditions
- To see some cool, recently launched products produced by ICPSR that your patrons may have interest in using

Before Starting

Definitions

User Testing: the process of collecting research data from real users while they are using a digital product whether a prototype or fully launched

User: an individual that uses or has the propensity to use a digital product; includes the individuals who actually interact with the product as well as those directing/training users to interact with the product

User Support: individuals who assist hundreds of users annually who are interacting with a product whether support is for general questions or to solve problems

A Note of Digital Data Gratitude!

Several of ICPSR's recent digital data products (indicated with Research Data Ecosystem) have been made possible with resources provided by the U.S. National Science Foundation.

The Research Data Ecosystem: a National Resource for Reproducible, Robust, and Transparent Social Science Research in the 21st Century, is funded by the National Science Foundation as a Mid-scale RI-2 Infrastructure Award (#1946932).

U.S. National Science Foundation

User Research: The Practicalities

Purpose: Observe, log, and analyze user interactions with a product, noting successes, challenges, and failures in order to improve user experience on the front-end and content received in the back-office

Challenges:

- Finding & recruiting "real" users
- Expense $$ and time intensive
- Live user research is a lab experience; users often act differently in the wild!
- Users represent one experience — their own! Is their experience unique or common — that is the question!

Process: At-A-Glance

1st Step
Define the problem

What is the (user) problem(s) to solve? How will it help the producer of the product (it's okay to help yourself!)?

Hint: Key to success — include user support staff in all phases starting with problem definition.

2nd Step
Define the user

Who is the primary user? Secondary?

Hint: Define demographically, experience levels (substantive, technological, methodological)

3rd Step
Test, Research, Build, and, if needed, Pivot

- Recruit for testing and reward ($)
- Conduct research again and again and again
- Invite user support to help interpret research

Hint: Digital Products — Be prepared to pivot when new technology arrives (AI changes everything!)

Real-life Examples from ICPSR Digital Data Product Releases

Product Research Data Ecosystem
TurboCurator

Problem
Depositors not often knowledgeable on **creating Findable metadata** efficiently

Solution
ICPSR's TurboCurator — designed to semi-automate the process

Pivot
Technology-inspired: AI - ChatGPT

Identified three key metadata areas (Title, Description, Key Words)

- Depositor inputs general information about the study, dataset, or file
- TurboCurator, powered in part by AI, outputs accepted formats for title, description, and controlled key words that depositor can cut/paste to describe the data deposit

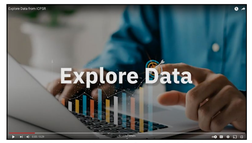
https://myumi.ch/AZwPe

Product Research Data Ecosystem
Explore Data

Problem
A need to "look into" the data to learn more about "what's in there" before downloading it or analyzing it; **make data more Accessible**

Solution
ICPSR's Explore Data

Pivot
User-inspired: Users advised against another statistical analysis tool; keep it simple



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Product Research Data Ecosystem
Researcher Passport
Authentication & Authorization System

Problem
A need for an updated authentication and authorization system to exit ICPSR from password management **(increase security) and make access to restricted-use data less cumbersome**

Solution
ICPSR's Researcher Passport — launching in two phases:

- Basic Passport — basic account for data access
- Enhanced Passport — account with profile information for access to restricted-use data

Pivot
User-inspired: Users did not want to enter/keep updated another profile nor set up passwords; ICPSR's authentication system built to use other platforms for authentication/authorization (InCommon, Google, LinkedIn, OrcID)

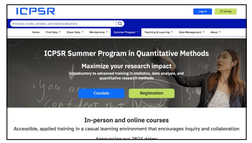
https://researcherpassport.icpsr.umich.edu/help

Product
ICPSR Summer Program Website

Problem
Website was hard to navigate and Google could not find it (needed to be optimized for Search Engine Optimization (SEO)

Solution
Design website to port users where they want to go quickly; properly title headings and subheadings to help search engines understand the offer and bring users to the page

Pivot
User-inspired: Reorganize and simplify navigation; rename courses for clearer distinction between 3-5-day workshops and 3-week sessions and using course terms that students use to search with



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